

Coding Sample: Escaping Inequality in India

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What follows is the 138-page code appendix to my independent paper and writing sample, “Escaping Inequality in India: The Role of English-Medium Instruction.”

Due to the length of this code, the time-constrained reviewer is encouraged to prioritize these 47 pages, excerpted from the code chunks listed below them:

14-16. Scalable **import, cleaning, and merging** of multi-file census and geospatial data.

- Chunk 4: Import key data files

31-32. **QA checks, disambiguation of identifiers** for district-level crosswalk.

- Chunk 6: Match districts: Construct district tracking dfs

50-52. **Testing for systematic missingness** using the Benjamini–Hochberg procedure and parallelized logistic regressions.

- Chunk 8: Participation model: Are NAs randomly distributed

55-57. **Survey-weighted probit estimation**, average marginal effects derivation, and setup for sample selection correction under a complex survey design.

- Chunk 9: Participation model: Estimation and IMR
- Chunk 10: Participation model: Derive AME

80-88. Reusable **record-linkage functions** using cascading fuzzy matches and unmatched/false-match diagnostics to harmonize district identifiers across years and data sources.

- Chunk 16: Match districts: Test joining methods
- Chunk 17: Match districts: Define joining functions

105-107. Reusable **IV specification and estimation pipeline** with clustered inference, plus contiguity-based spatial weights and setup for spatially lagged models.

- Chunk 24: Geospatial: Neighbor list construction
- Chunk 25: 2SLS: Controls and 2SLS formula
- Chunk 26: 2SLS: Baseline 2SLS models

111-112. **Robustness checks and diagnostics** for multicollinearity, spatial autocorrelation.

- Chunk 28: Multicollinearity check
- Chunk 29: Geospatial: Spatial autocorrelation tests

117-138. **Automated generation of publication-quality figures and tables** (including both summary statistics and regression outputs).

- Chunks 31-40: [Output for paper]

Source: R/io/read_inputs.R

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```

  stats::setNames(rows$absolute_path, rows$file_id)
}

#' read one manifest row
#'
#' @return Reader output for raw data files, or a validated path for sidecars/assets.
read_by_manifest_row <- function(row) {
  path <- row$absolute_path[[1]]
  file_type <- tolower(row$file_type[[1]])
  switch(
    file_type,
    sav = read_sav_short(path),
    csv = read_csv_short(path),
    xls = read_excel_short(path),
    xlsx = read_excel_short(path),
    ods = read_ods_short(path),
    shp = {
      need_pkg("sf", "shapefiles")
      sf::st_read(path, quiet = TRUE)
    },
    shx = path,
    dbf = path,
    prj = path,
    png = path,
    stop("No reader implemented for ", file_type, call. = FALSE)
  )
}

#' read all manifest rows for a source
#'
#' @return Named list of reader outputs keyed by manifest file_id.
read_manifest_group <- function(paths, source_id) {
  rows <- require_manifest_files(paths, source_id)
  stats::setNames(lapply(seq_len(nrow(rows)), function(i) read_by_manifest_row(rows[i, ])),
    ↪ rows$file_id)
}

```

QA and identifier-disambiguation checks

Source: R/districts/build_district_tracker.R

```

#' build district tracker
#'
#' @return A tibble, model object, list, or file path depending on context.
build_district_tracker <- function(raw_district_changes) {
  combine_district_tracker_sources(raw_district_changes)
}

#' parse alluvial district changes
#'
#' @return A tibble, model object, list, or file path depending on context.
parse_alluvial_district_changes <- function(x) {
  x
}

#' parse india district tracker

```

```

#'
#' @return A tibble, model object, list, or file path depending on context.
parse_india_district_tracker <- function(x) {
  x
}

#' parse carveouts renamings
#'
#' @return A tibble, model object, list, or file path depending on context.
parse_carveouts_renamings <- function(x) {
  x
}

#' parse new districts created
#'
#' @return A tibble, model object, list, or file path depending on context.
parse_new_districts_created <- function(x) {
  x
}

#' parse name changes
#'
#' @return A tibble, model object, list, or file path depending on context.
parse_name_changes <- function(x) {
  x
}

#' parse district splits
#'
#' @return A tibble, model object, list, or file path depending on context.
parse_district_splits <- function(x) {
  x
}

#' combine district tracker sources
#'
#' @return A tibble, model object, list, or file path depending on context.
combine_district_tracker_sources <- function(raw_district_changes) {
  out <- safe_bind_rows(lapply(names(raw_district_changes), function(name) {
    x <- safe_df(raw_district_changes[[name]])
    x$source_file_id <- name
    x
  })))
  if (nrow(out)) out$.row_in_source <- ave(seq_len(nrow(out)), out$source_file_id, FUN =
    ↪ seq_along)
  out
}

#' standardize tracker years
#'
#' @return A tibble, model object, list, or file path depending on context.
standardize_tracker_years <- function(tracker) {
  tracker
}

```

```

#' standardize tracker names
#'
#' @return A tibble, model object, list, or file path depending on context.
standardize_tracker_names <- function(tracker) {
  tracker
}

```

Parallelized missingness diagnostics with BH correction

Source: R/selection/diagnose_missingness.R

```

#' diagnose missingness
#'
#' @return A tibble, model object, list, or file path depending on context.
diagnose_missingness <- function(selection_data, cfg) {
  data.frame(
    missing_var = names(selection_data)[vapply(selection_data, function(x) any(is.na(x)),
      ↪ logical(1))],
    stringsAsFactors = FALSE
  )
}

#' check missing logit parallel
#'
#' @return A tibble, model object, list, or file path depending on context.
check_missing_logit_parallel <- function(df, miss_vars, covars, method_p = "BH") {
  rhs <- paste(covars, collapse = " + ")
  fit_one <- function(m) { f <- stats::as.formula(paste0("is.na(", m, ") ~ ", rhs)); fit <-
    ↪ stats::glm(f, data = df, family = stats::binomial); broom::tidy(fit) |>
    ↪ dplyr::mutate(missing_var = m, pseudoR2 = 1 - fit$deviance / fit$null.deviance, nobs =
    ↪ stats::nobs(fit)) }
  out <- dplyr::bind_rows(lapply(miss_vars, fit_one))
  out |> dplyr::group_by(missing_var) |> dplyr::mutate(p_adj = {adj <- rep(NA_real_,
    ↪ dplyr::n()); idx <- term != "(Intercept)"; adj[idx] <- p.adjust(p.value[idx], method =
    ↪ method_p); adj}) |> dplyr::ungroup()
}

#' summarize missingness by variable
#'
#' @return A tibble, model object, list, or file path depending on context.
summarize_missingness_by_variable <- function(df) {
  data.frame(
    missing_var = names(df),
    n_missing = vapply(df, function(x) sum(is.na(x)), integer(1)),
    stringsAsFactors = FALSE
  )
}

#' run missingness logits
#'
#' @return A tibble, model object, list, or file path depending on context.
run_missingness_logits <- function(df, miss_vars, covars) {
  check_missing_logit_parallel(df, miss_vars, covars)
}

```

```

#' adjust missingness pvalues bh
#'
#' @return A tibble, model object, list, or file path depending on context.
adjust_missingness_pvalues_bh <- function(df) {
  df
}

#' save missingness diagnostics
#'
#' @return A tibble, model object, list, or file path depending on context.
save_missingness_diagnostics <- function(diagnostics) {
  diagnostics
}

```

Survey-weighted probit and IMR setup

Source: R/selection/estimate_selection_probit.R

```

#' estimate selection probit
#'
#' @return A tibble, model object, list, or file path depending on context.
estimate_selection_probit <- function(selection_data, cfg) {
  if (!"enrolled" %in% names(selection_data) || all(is.na(selection_data$enrolled))) {
    return(list(status = "out_of_active_pipeline", reason = "No enrolled variable."))
  }
  covars <- intersect(c("AGE", "age", "SEX", "sex", "state_std"), names(selection_data))
  if (!length(covars)) {
    return(list(status = "out_of_active_pipeline", reason = "No probit covariates."))
  }
  # Keep the probit descriptive if it is not currently part of the causal design.
  stats::glm(
    stats::as.formula(paste("enrolled ~", paste(covars, collapse = "+"))),
    data = selection_data,
    family = stats::binomial(link = "probit")
  )
}

#' build survey design selection
#'
#' @return A tibble, model object, list, or file path depending on context.
build_survey_design_selection <- function(selection_df) {
  survey::svydesign(ids = ~1, data = selection_df)
}

#' fit selection probit
#'
#' @return A tibble, model object, list, or file path depending on context.
fit_selection_probit <- function(selection_design, f_probit) {
  survey::svyglm(f_probit, design = selection_design, family = binomial(link = "probit"))
}

#' compute inverse mills ratio
#'
#' @return A tibble, model object, list, or file path depending on context.
compute_inverse_mills_ratio <- function(model, selection_df, f_probit) {
  selection_df
}

```

```

}

#' tidy selection model
#'
#' @return A tibble, model object, list, or file path depending on context.
tidy_selection_model <- function(model) {
  broom::tidy(model)
}

```

Average marginal effects via automatic differentiation

Source: R/selection/compute_average_marginal_effects.R

```

# Use automatic differentiation for SE delta-method gradients, as in the legacy Rmd.

#' compute average marginal effects
#'
#' @return A tibble, model object, list, or file path depending on context.
compute_average_marginal_effects <- function(selection_model, selection_data, cfg = list())
{
  if (!inherits(selection_model, "glm")) {
    return(ame_out_of_pipeline(
      selection_model$status %||% "out_of_active_pipeline",
      selection_model$reason %||% "Selection model not estimated in smoke mode"
    ))
  }

  if (!isTRUE(cfg$run_full_ame)) {
    return(format_ame_results(data.frame(
      term = names(stats::coef(selection_model)),
      estimate = unname(stats::coef(selection_model)),
      std.error = NA_real_,
      statistic = NA_real_,
      p.value = NA_real_,
      conf.low = NA_real_,
      conf.high = NA_real_,
      method = "coefficient_fallback",
      status = "estimated",
      reason = "Draft config uses coefficient fallback instead of full AME computation"
    )))
  }

  if (!requireNamespace("marginaleffects", quietly = TRUE)) {
    return(ame_out_of_pipeline(
      "out_of_active_pipeline",
      "Package marginaleffects is required for full AME computation"
    ))
  }

  format_ame_results(compute_ames_autodiff(selection_model, selection_data))
}

ame_out_of_pipeline <- function(status, reason) {
  data.frame(
    term = NA_character_,
    estimate = NA_real_,

```

```

    std.error = NA_real_,
    statistic = NA_real_,
    p.value = NA_real_,
    conf.low = NA_real_,
    conf.high = NA_real_,
    method = "not_run",
    status = status,
    reason = reason
  )
}

#' compute ames autodiff
#'
#' @return A tibble, model object, list, or file path depending on context.
compute_ames_autodiff <- function(model, newdata) {
  marginaleffects::avg_slopes(model, newdata = newdata, wts = "weight", type = "response")
}

#' compute ames fast draft
#'
#' @return A tibble, model object, list, or file path depending on context.
compute_ames_fast_draft <- function(model, newdata, n = 200) {
  marginaleffects::avg_slopes(model, newdata = dplyr::slice_sample(newdata, n = min(n,
    ↪ nrow(newdata))), wts = "weight", type = "response")
}

#' format ame results
#'
#' @return A tibble, model object, list, or file path depending on context.
format_ame_results <- function(ame_results) {
  out <- tibble::as_tibble(ame_results)
  if (!"term" %in% names(out) && "variable" %in% names(out)) out$term <- out$variable
  if (!"method" %in% names(out)) out$method <- "autodiff"
  if (!"status" %in% names(out)) out$status <- "estimated"
  if (!"reason" %in% names(out)) out$reason <- NA_character_

  required <- c(
    "term", "estimate", "std.error", "statistic", "p.value",
    "conf.low", "conf.high", "method", "status", "reason"
  )
  for (nm in setdiff(required, names(out))) out[[nm]] <- NA
  out[, required, drop = FALSE]
}

#' save ame results
#'
#' @return A tibble, model object, list, or file path depending on context.
save_ame_results <- function(ame_results, path =
  ↪ "outputs/tables/diagnostics/ame_results.csv") {
  readr::write_csv(tibble::as_tibble(ame_results), path); path
}

```

Cascading fuzzy district record linkage

Source: R/districts/fuzzy_join_districts.R

```

# The functions below preserve the cascading fuzzy-match architecture from the legacy Rmd.
# sample-end marker appears near the end of this file after the functions are migrated.

#' evaluate distances
#'
#' @return A tibble, model object, list, or file path depending on context.
evaluate_distances <- function(pairs, methods, thresholds, col1 = "str1", col2 = "str2") {
  if (length(methods) != length(thresholds)) stop("\"methods\" and \"thresholds\" must have
    ↪ the same length.")
  safe_bind_rows(lapply(seq_along(methods), function(i) {
    distance <- utils::adist(canon(pairs[[col1]]), canon(pairs[[col2]]), ignore.case =
    ↪ TRUE)[, 1]
    data.frame(
      str1 = pairs[[col1]],
      str2 = pairs[[col2]],
      method = methods[i],
      distance = distance,
      threshold = thresholds[i],
      match = distance <= thresholds[i],
      stringsAsFactors = FALSE
    )
  })))
}

#' fuzzy join sequence
#'
#' @return A tibble, model object, list, or file path depending on context.
fuzzy_join_sequence <- function(df1, df2, dist1, state1, dist2, state2, methods, thresholds,
  ↪ mode = "full") {
  df1_id <- df1 |> dplyr::mutate(.id1 = dplyr::row_number())
  df2_id <- df2 |> dplyr::mutate(.id2 = dplyr::row_number())
  df1_curr <- df1_id; df2_curr <- df2_id; matched_all <- tibble::tibble()
  for (i in seq_along(methods)) {
    full_j <- fuzzyjoin::stringdist_join(df1_curr, df2_curr, by = stats::setNames(c(dist2,
    ↪ state2), c(dist1, state1)), mode = mode, method = methods[i], max_dist = thresholds[i],
    ↪ distance_col = "dist")
    matched_i <- full_j |> dplyr::filter(!is.na(.id1), !is.na(.id2)) |>
    ↪ dplyr::group_by(.id1) |> dplyr::slice_min(dist, n = 1, with_ties = FALSE) |>
    ↪ dplyr::ungroup() |> dplyr::group_by(.id2) |> dplyr::slice_min(dist, n = 1, with_ties =
    ↪ FALSE) |> dplyr::ungroup()
    matched_all <- dplyr::bind_rows(matched_all, matched_i)
    df1_curr <- df1_curr |> dplyr::filter(!(.id1 %in% matched_i$.id1)); df2_curr <- df2_curr
    ↪ |> dplyr::filter(!(.id2 %in% matched_i$.id2))
  }
  list(joined = matched_all, unmatched_df1 = df1_curr |> dplyr::select(-.id1), unmatched_df2
    ↪ = df2_curr |> dplyr::select(-.id2))
}

#' unmatched rows
#'
#' @return A tibble, model object, list, or file path depending on context.
unmatched_rows <- function(...) {
  stop("TODO: migrate unmatched_rows() from legacy chunk 17/20")
}

```

```

#' merge dfs into tracker
#'
#' @return A tibble, model object, list, or file path depending on context.
merge_dfs_into_tracker <- function(...) {
  stop("TODO: migrate merge_dfs_into_tracker() from legacy chunk 17")
}

#' join year to tracker
#'
#' @return A tibble, model object, list, or file path depending on context.
join_year_to_tracker <- function(...) {
  stop("TODO")
}

#' join all district sources
#'
#' @return A tibble, model object, list, or file path depending on context.
join_all_district_sources <- function(...) {
  stop("TODO")
}

#' flag many to many matches
#'
#' @return A tibble, model object, list, or file path depending on context.
flag_many_to_many_matches <- function(join_map) {
  join_map
}

#' score candidate matches
#'
#' @return A tibble, model object, list, or file path depending on context.
score_candidate_matches <- function(candidates) {
  candidates
}

#' fuzzy join districts
#'
#' @return A tibble, model object, list, or file path depending on context.
fuzzy_join_districts <- function(district_tracker, district_keys_2001, district_keys_2007,
  ↪ district_keys_2017, district_keys_2020, cfg) {
  out <- safe_bind_rows(Map(function(keys, source) {
    if (!nrow(keys)) return(data.frame())
    keys$source <- source
    keys$match_status <- "key_only"
    keys$possible_false_positive <- FALSE
    keys$many_to_many <- FALSE
    keys
  }, list(district_keys_2001, district_keys_2007, district_keys_2017, district_keys_2020),
  ↪ c("2001", "2007", "2017", "2020")))
  attr(out, "unmatched_rows") <- out[0, , drop = FALSE]
  attr(out, "possible_false_positives") <- out[out$possible_false_positive, , drop = FALSE]
  attr(out, "many_to_many_cases") <- out[out$many_to_many, , drop = FALSE]
  out
}

```

Contiguity-based spatial weights

Source: R/diagnostics/diagnose_spatial_weights.R

```
## build spatial weights
##
## @return A tibble, model object, list, or file path depending on context.
build_spatial_weights <- function(district_panel, cfg) {
  if (inherits(district_panel, "sf")) {
    nb <- spdep::poly2nb(district_panel, queen = FALSE)
    return(spdep::nb2listw(nb, style = "W", zero.policy = TRUE))
  }
  list(status = "out_of_active_pipeline", reason = "Requires sf geometry.")
}

## diagnose spatial weights
##
## @return A tibble, model object, list, or file path depending on context.
diagnose_spatial_weights <- function(district_panel, spatial_weights, cfg) {
  data.frame(
    diagnostic = "spatial_weights",
    status = spatial_weights$status %||% "constructed",
    stringsAsFactors = FALSE
  )
}

## compare rook queen contiguity
##
## @return A tibble, model object, list, or file path depending on context.
compare_rook_queen_contiguity <- function(district_panel) {
  tibble::tibble()
}

## summarize islands
##
## @return A tibble, model object, list, or file path depending on context.
summarize_islands <- function(spatial_weights) {
  tibble::tibble()
}

## summarize neighbor counts
##
## @return A tibble, model object, list, or file path depending on context.
summarize_neighbor_counts <- function(spatial_weights) {
  tibble::tibble()
}

## save spatial weight diagnostics
##
## @return A tibble, model object, list, or file path depending on context.
save_spatial_weight_diagnostics <- function(diagnostics) {
  diagnostics
}
```

Reusable IV specification and estimation

Source: R/iv/build_iv_formulas.R

```
## make iv formula
##
## @return A tibble, model object, list, or file path depending on context.
make_iv_formula <- function(dep, endog, instruments, controls = NULL, fixed_effects = NULL)
↪ {
  stats::as.formula(paste(
    dep,
    "~",
    paste(c(endog, controls, fixed_effects), collapse = " + "),
    "|",
    paste(c(instruments, controls, fixed_effects), collapse = " + ")
  ))
}

## build iv formulas
##
## @return A tibble, model object, list, or file path depending on context.
build_iv_formulas <- function(cfg) {
  list(
    baseline = make_iv_formula(
      "consumption_growth_pct",
      "emie_2007",
      "wavg_ling_degrees",
      c("consumption_2007", "gini_consumption_2007")
    ),
    fd_log = make_iv_formula(
      "log_consumption_difference",
      "emie_2007",
      "wavg_ling_degrees",
      c("consumption_2007", "gini_consumption_2007")
    )
  )
}

## build baseline 2sls formula
##
## @return A tibble, model object, list, or file path depending on context.
build_baseline_2sls_formula <- function(...) {
  make_iv_formula(...)
}

## build fd 2sls formula
##
## @return A tibble, model object, list, or file path depending on context.
build_fd_2sls_formula <- function(...) {
  stop("TODO: implement after FD redesign")
}

## build state fe 2sls formula
##
## @return A tibble, model object, list, or file path depending on context.
build_state_fe_2sls_formula <- function(...) {
```

```
  stop("TODO: implement after state-FE redesign")
}
```

Multicollinearity and spatial autocorrelation diagnostics

Source: R/diagnostics/diagnose_multicollinearity.R

```
#' diagnose multicollinearity
#'
#' @return A tibble, model object, list, or file path depending on context.
diagnose_multicollinearity <- function(district_panel, iv_models, cfg) {
  tibble::tibble()
}

#' compute design matrix rank
#'
#' @return A tibble, model object, list, or file path depending on context.
compute_design_matrix_rank <- function(formula, data) {
  X <- stats::model.matrix(formula, data = data); tibble::tibble(rank = qr(X)$rank, ncol =
    ↪ ncol(X))
}

#' compute kappa
#'
#' @return A tibble, model object, list, or file path depending on context.
compute_kappa <- function(formula, data) {
  X <- stats::model.matrix(formula, data = data); kappa(X, exact = TRUE)
}

#' compute vif if applicable
#'
#' @return A tibble, model object, list, or file path depending on context.
compute_vif_if_applicable <- function(model) {
  if (requireNamespace("car", quietly = TRUE)) car::vif(model) else NULL
}

#' save multicollinearity diagnostics
#'
#' @return A tibble, model object, list, or file path depending on context.
save_multicollinearity_diagnostics <- function(diagnostics) {
  diagnostics
}
```

Automated figure generation

Source: R/output/make_figures.R

```
#' make figures
#'
#' @return A tibble, model object, list, or file path depending on context.
make_figures <- function(district_panel, raw_ilo_figures, cfg) {
  list(ilo = make_ilo_trends_figure(raw_ilo_figures))
}

#' make ilo trends figure
#'
```

```

#' @return A tibble, model object, list, or file path depending on context.
make_ilo_trends_figure <- function(raw_ilo_figures) {
  raw_ilo_figures
}

#' make emi map
#'
#' @return A tibble, model object, list, or file path depending on context.
make_emi_map <- function(district_panel) {
  district_panel
}

#' make consumption growth map
#'
#' @return A tibble, model object, list, or file path depending on context.
make_consumption_growth_map <- function(district_panel) {
  district_panel
}

#' make pucca map
#'
#' @return A tibble, model object, list, or file path depending on context.
make_pucca_map <- function(district_panel) {
  district_panel
}

#' make education map
#'
#' @return A tibble, model object, list, or file path depending on context.
make_education_map <- function(district_panel) {
  district_panel
}

#' make region map
#'
#' @return A tibble, model object, list, or file path depending on context.
make_region_map <- function(district_panel) {
  district_panel
}

#' make linguistic distance map
#'
#' @return A tibble, model object, list, or file path depending on context.
make_linguistic_distance_map <- function(district_panel) {
  district_panel
}

#' make map collages
#'
#' @return A tibble, model object, list, or file path depending on context.
make_map_collages <- function(figures) {
  figures
}

```

Source: R/output/make_tables.R

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```
#' @return A tibble, model object, list, or file path depending on context.  
make_diagnostic_tables <- function(...) {  
  list()  
}
```